IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

Executive Summary

Predicting the successful landing of SpaceX Falcon 9 rocket first stages

- Methodologies
  - Data collection from various sources
  - Exploratory data analysis (EDA)
  - Use machine learning to predict outcome
- Summary of all results
  - Achieve 89% accuracy with decision tree classifier
  - Payload mass and orbit type are significant factors

Introduction

- **Background**
  - SpaceX's Falcon 9 rocket is renowned for its reusability, which significantly reduces space launch costs. The first stage landing success is crucial for cost savings, as successful landings allow for rocket reuse. This project aims to predict whether the Falcon 9 first stage will land successfully based on historical launch data.
- **Problem statement**
  - Determine the probability of a successful first stage landing to help competing companies bid against SpaceX for rocket launches. The binary classification problem uses features like payload, orbit, launch site, and booster version.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - SpaceX REST API for rocket, launch, and core details.
  - Web scraping Wikipedia pages for historical Falcon 9 launch records.
- Perform data wrangling
  - Handled missing values, one-hot encoding, normalize data
  - Final dataset shape: Approximately 90 launches with 18 features after processing.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Try different models: Logistic Regression, Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (KNN)
  - Hyperparameter tuning via GridSearchCV

Data Collection

Data was gathered from two main sources:

- SpaceX REST API for rocket, launch, and core details.
- Web scraping Wikipedia pages for historical Falcon 9 launch records.

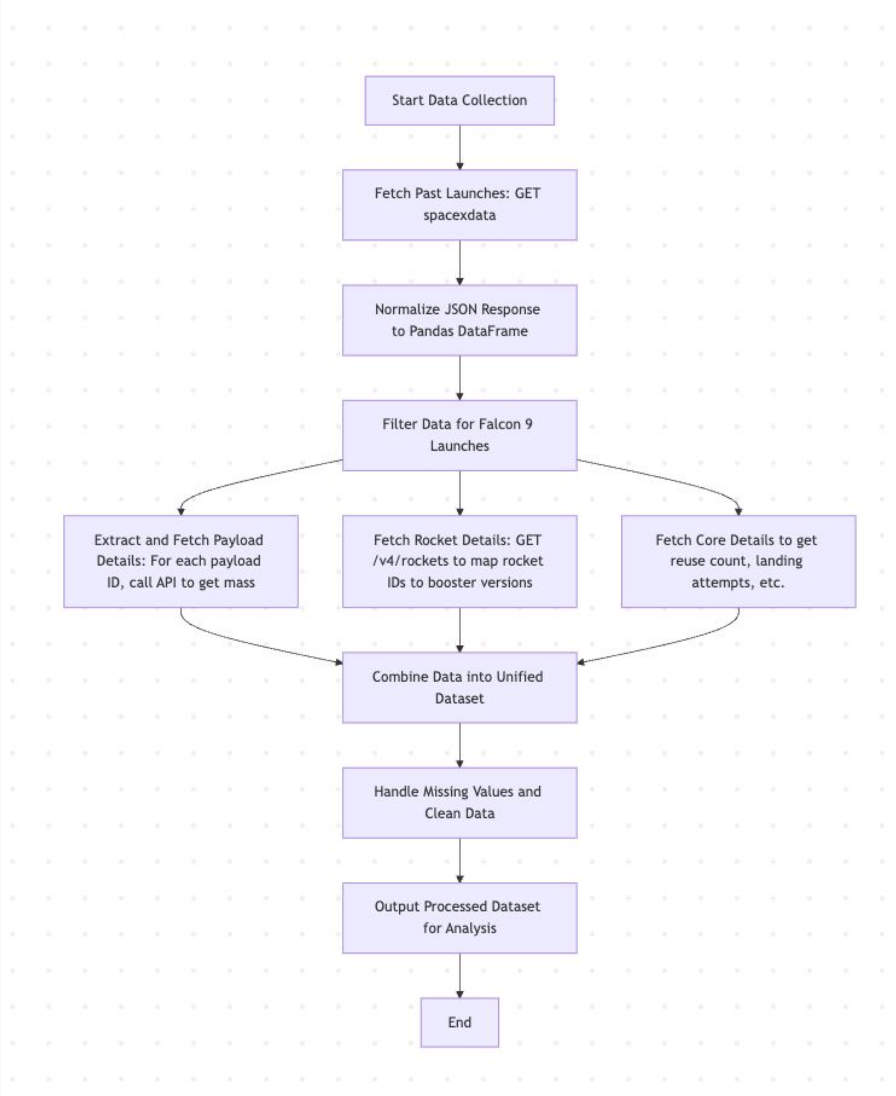
The dataset includes variables such as:

- Flight number
- Payload mass
- Orbit type
- Launch site
- Landing outcome (success/failure)
- Booster version
- Number of flights

Data Collection – SpaceX API

- Data collection with SpaceX REST calls using key phrases and flowcharts

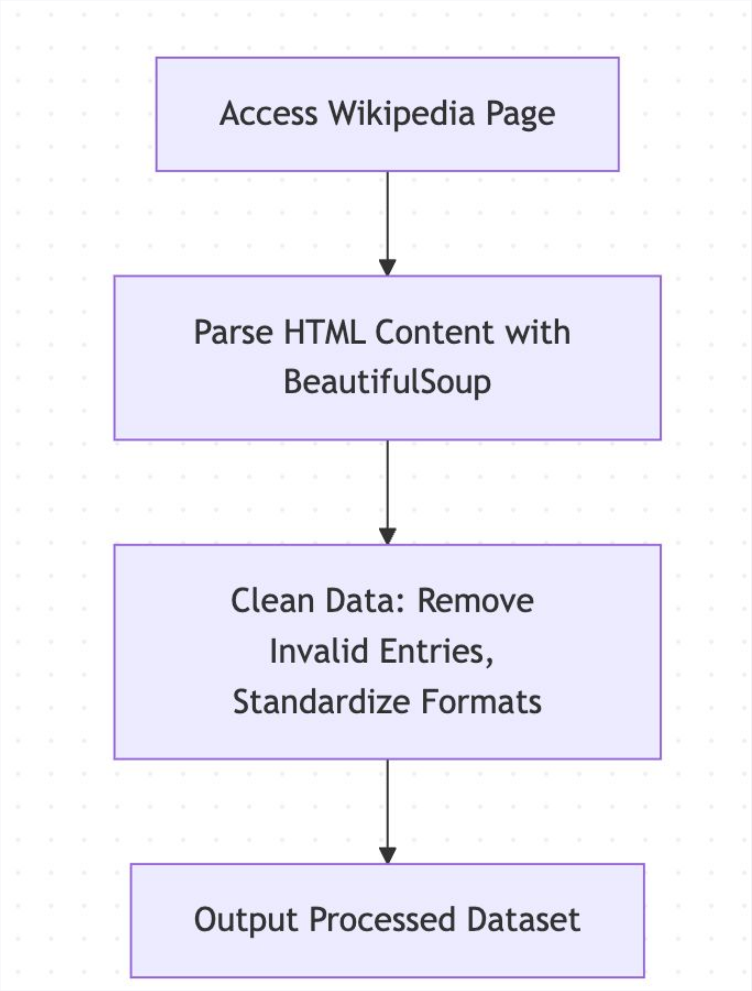
- Notebook: https://github.com/ittus/learn-data-science/blob/main/capstone/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts

- Notebook: https://github.com/ittus/learn-data-science/blob/main/capstone/jupyter-labs-webscraping.ipynb



Data Wrangling

- Handled missing values (e.g., filled null payload masses with mean values).
- Converted categorical variables (e.g., orbit, launch site) to numerical using one-hot encoding.
- Standardized numerical features for model input.
- Final dataset shape: Approximately 90 launches with 18 features after processing

EDA with Data Visualization

- Visualized success rates by launch site using bar charts (e.g., CCAFS SLC 40 had ~60% success).
- Scatter plots showed relationships between payload mass and orbit type with landing success.
- Interactive Folium maps marked launch sites and success/failure locations.
- Used Seaborn for correlation heatmaps to identify key predictors.

- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose

Notebook: https://github.com/ittus/learn-data-science/blob/main/capstone/jupyter-labs-eda-sql-coursera_sqllite.ipynb

EDA with SQL

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Notebook:
https://github.com/ittus/learn-data-science/blob/main/capstone/edadataviz.ipynb

Build an Interactive Map with Folium

- Create markers to show location of launch site
- Check relationship between launch site and success rate
- Notebook: https://github.com/ittus/learn-data-science/blob/main/capstone/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- Visualize to obtain information
    - Which site has the largest successful launches?
    - Which site has the highest launch success rate?
    - Which payload range(s) has the highest launch success rate?
    - Which payload range(s) has the lowest launch success rate?
    - Which F9 Booster version (v1.0, v1.1, FT, B4, B5, etc.) has the highest
    - launch success rate?
- Notebook: https://github.com/ittus/learn-data-science/blob/main/capstone/week3_plotly.py

Predictive Analysis (Classification)

- Split data into 80/20 train/test sets.
- Models trained: Logistic Regression, Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (KNN).
- Hyperparameter tuning via GridSearchCV (e.g., for tree depth in Decision Tree).
- Evaluation metrics: Accuracy, Precision, Recall, F1-Score, Confusion Matrix.

Notebook:
https://github.com/ittus/learn-data-science/blob/main/capstone/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

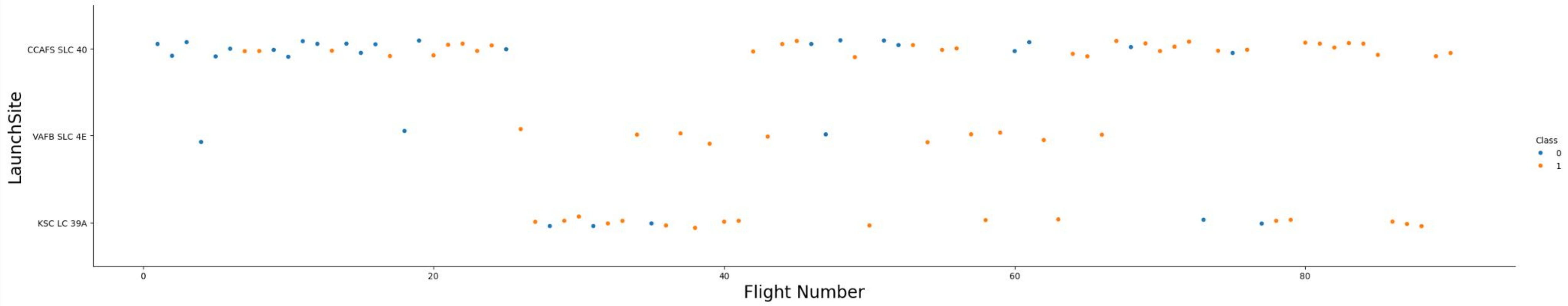
- Decision Tree performed best with 89% accuracy.
- Confusion matrices showed low false negatives, important for risk assessment.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.78	0.80	0.75	0.77
SVM	0.83	0.85	0.80	0.82
Decision Tree	0.89	0.90	0.85	0.87
KNN	0.83	0.82	0.80	0.81

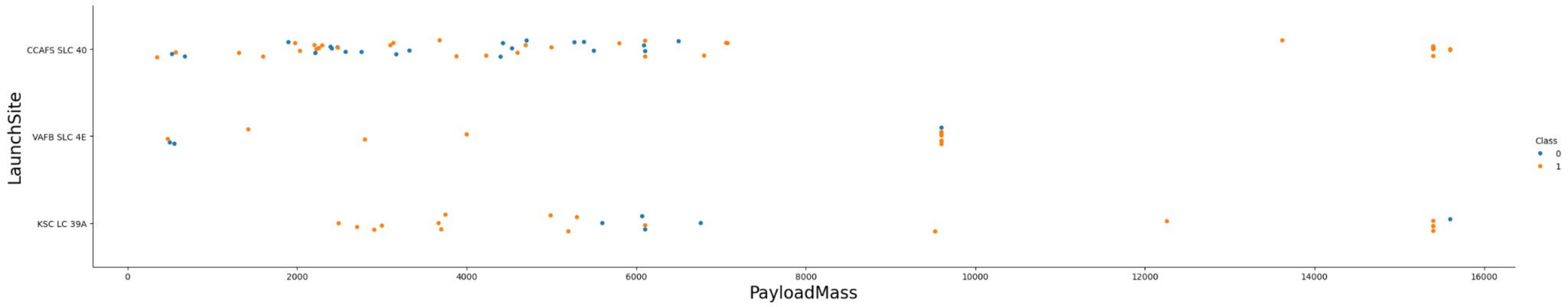
Section 2

Insights drawn from EDA

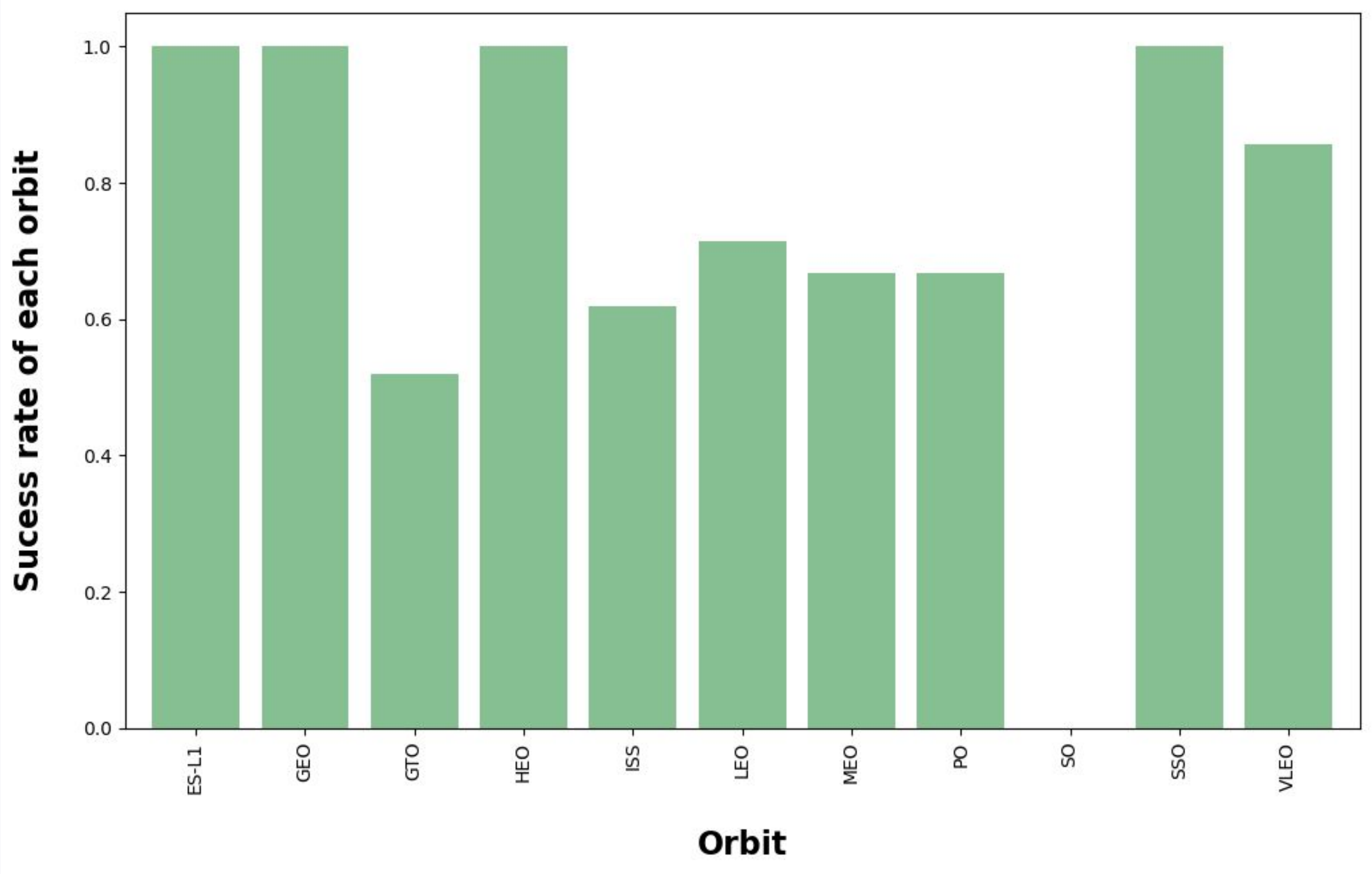
Flight Number vs. Launch Site



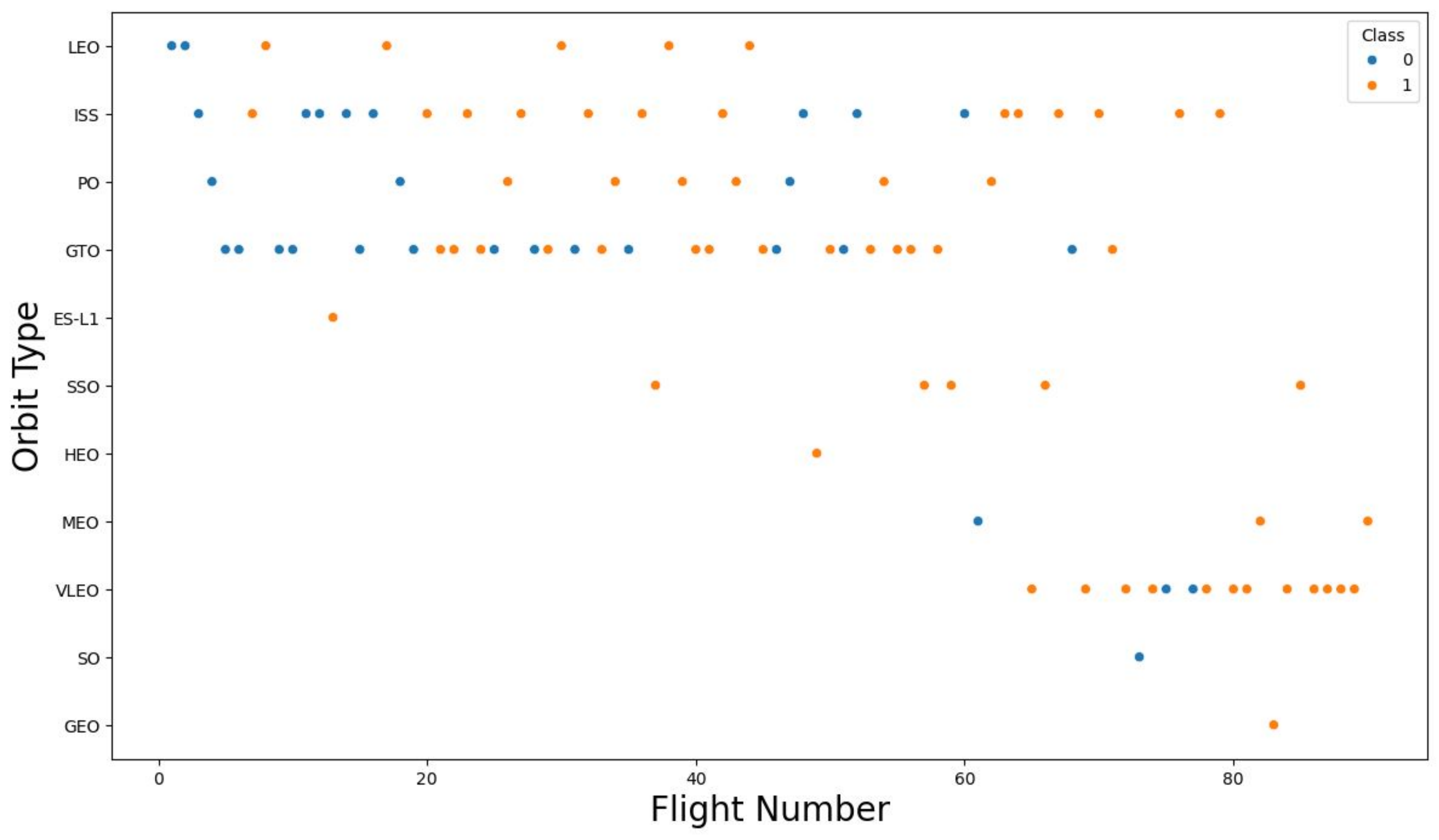
Payload vs. Launch Site



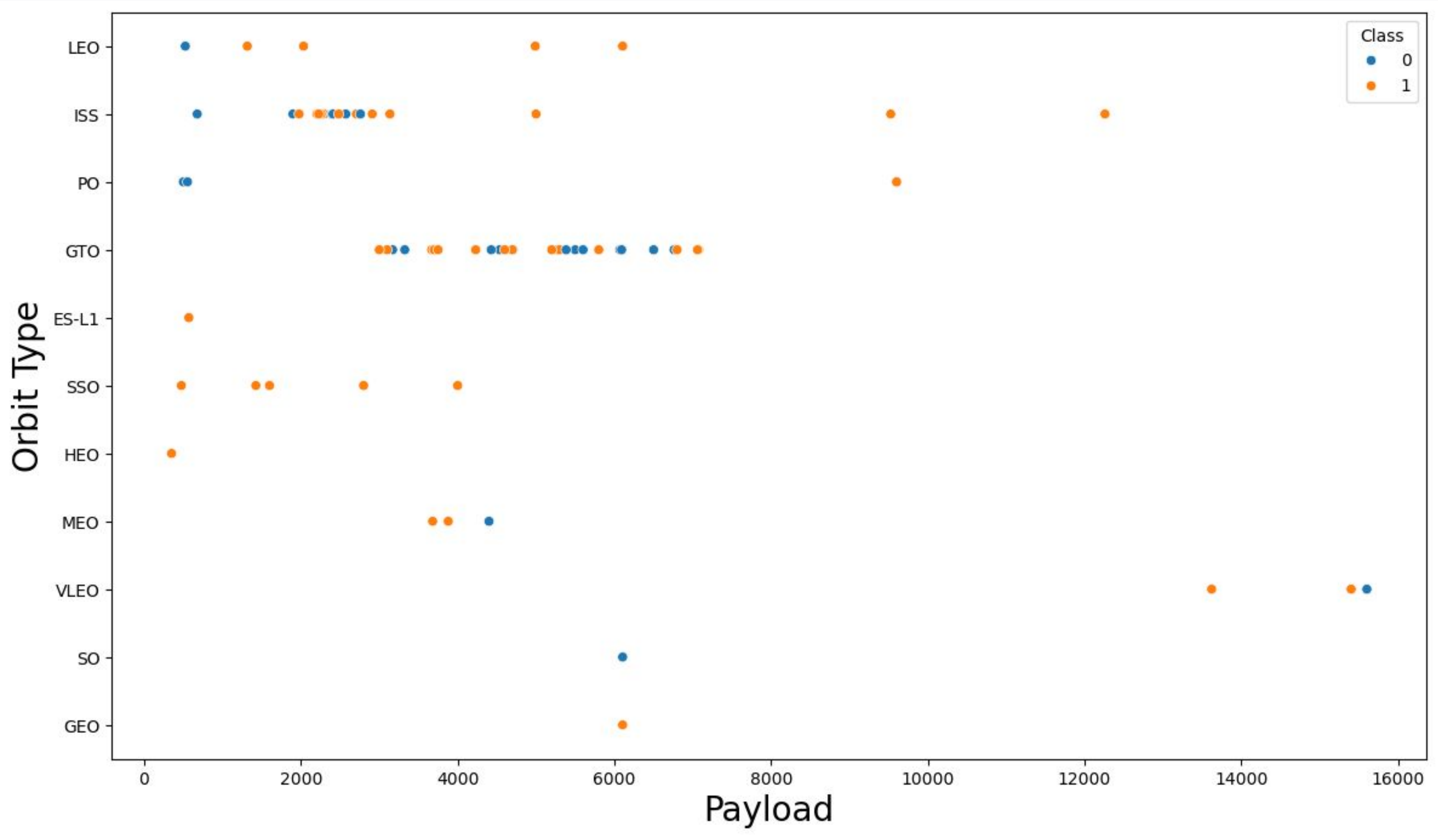
Success Rate vs. Orbit Type



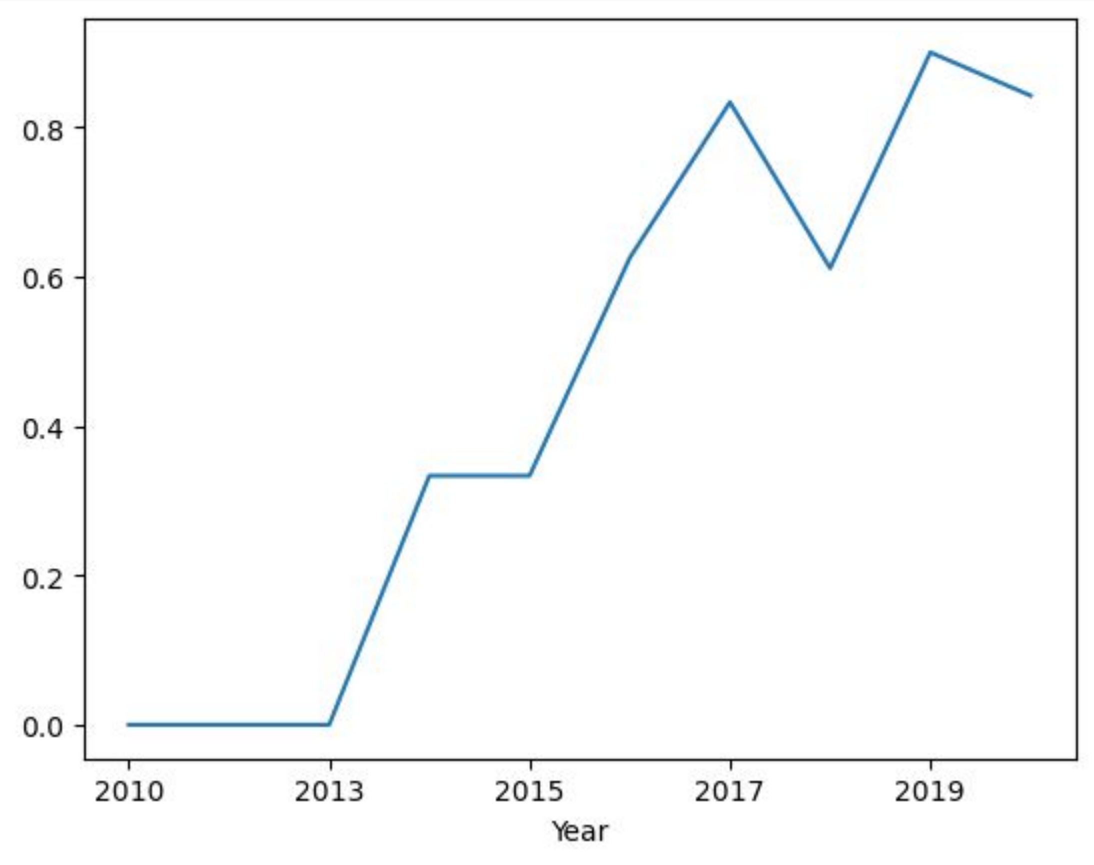
Flight Number vs. Orbit Type



Payload vs. Orbit Type



Launch Success Yearly Trend



All Launch Site Names

```
%sql SELECT DISTINCT LAUNCH_SITE FROM SPACEXTBL ORDER BY 1;
```

```
* sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
CCAFS SLC-40
KSC LC-39A
VAFB SLC-4E

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM SPACEXTBL WHERE LAUNCH_SITE LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (p
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (p
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	N
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	N
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	N

Total Payload Mass

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) AS TOTAL_PAYLOAD FROM SPACEXTBL WHERE PAYLOAD LIKE '%CRS%';
```

```
 * sqlite:///my_data1.db
Done.
```

TOTAL_PAYLOAD
111268

Average Payload Mass by F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) AS AVG_PAYLOAD FROM SPACEXTBL WHERE BOOSTER_VERSION = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

AVG_PAYLOAD
2928.4

First Successful Ground Landing Date

```
%sql SELECT MIN("DATE") FROM SPACEXTBL WHERE Landing_Outcome LIKE '%Success%'
```

```
 * sqlite:///my_data1.db
Done.
```

MIN("DATE")
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

```
%sql SELECT DISTINCT BOOSTER_VERSION FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000 AND Landing_Out
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

```
%sql SELECT MISSION_OUTCOME, COUNT(*) AS QTY FROM SPACEXTBL GROUP BY MISSION_OUTCOME ORDER BY MISSION_OUTCOME;
```

```
* sqlite:///my_data1.db
Done.
```

Mission_Outcome	QTY
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

```
%sql SELECT DISTINCT BOOSTER_VERSION FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_)
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1048.4
F9 B5 B1048.5
F9 B5 B1049.4
F9 B5 B1049.5
F9 B5 B1049.7
F9 B5 B1051.3
F9 B5 B1051.4
F9 B5 B1051.6
F9 B5 B1056.4
F9 B5 B1058.3
F9 B5 B1060.2
F9 B5 B1060.3

Task 9

2015 Launch Records

```
%sql SELECT substr("DATE", 6, 2) AS MONTH, "BOOSTER_VERSION", "LAUNCH_SITE" FROM SPACEXTBL\
WHERE Landing_Outcome = 'Failure (drone ship)' and substr("DATE",0, 5) = '2015'
```

```
 * sqlite:///my_data1.db
Done.
```

MONTH	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
%sql SELECT "LANDING_OUTCOME", COUNT("LANDING_OUTCOME") FROM SPACEXTBL\
WHERE "DATE" >= '2010-06-04' and "DATE" <= '2017-03-20' and "LANDING_OUTCOME" LIKE '%Success%'\
GROUP BY "LANDING_OUTCOME" \
ORDER BY COUNT("LANDING_OUTCOME") DESC ;
```

```
 * sqlite:///my_data1.db
Done.
```

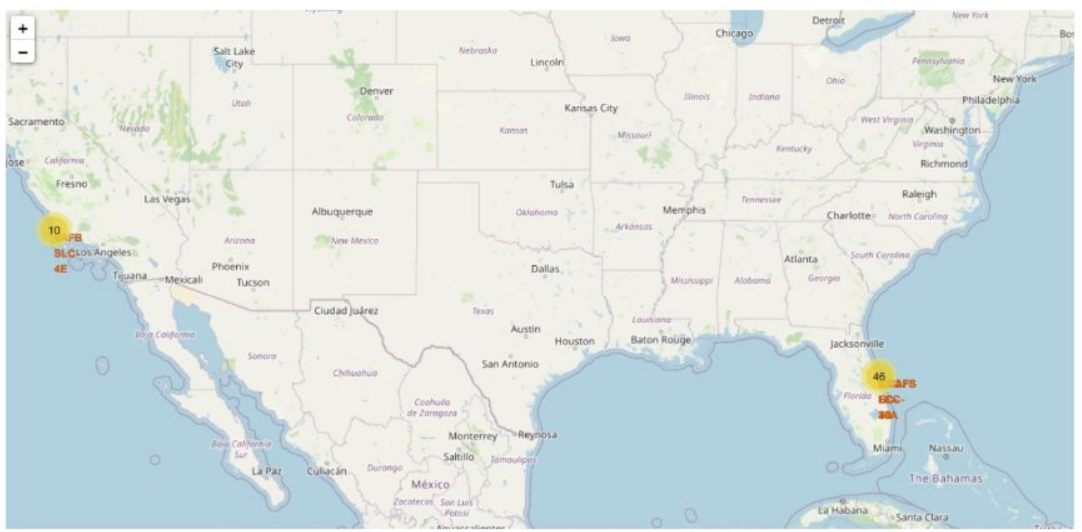
Landing_Outcome	COUNT("LANDING_OUTCOME")
Success (drone ship)	5
Success (ground pad)	3

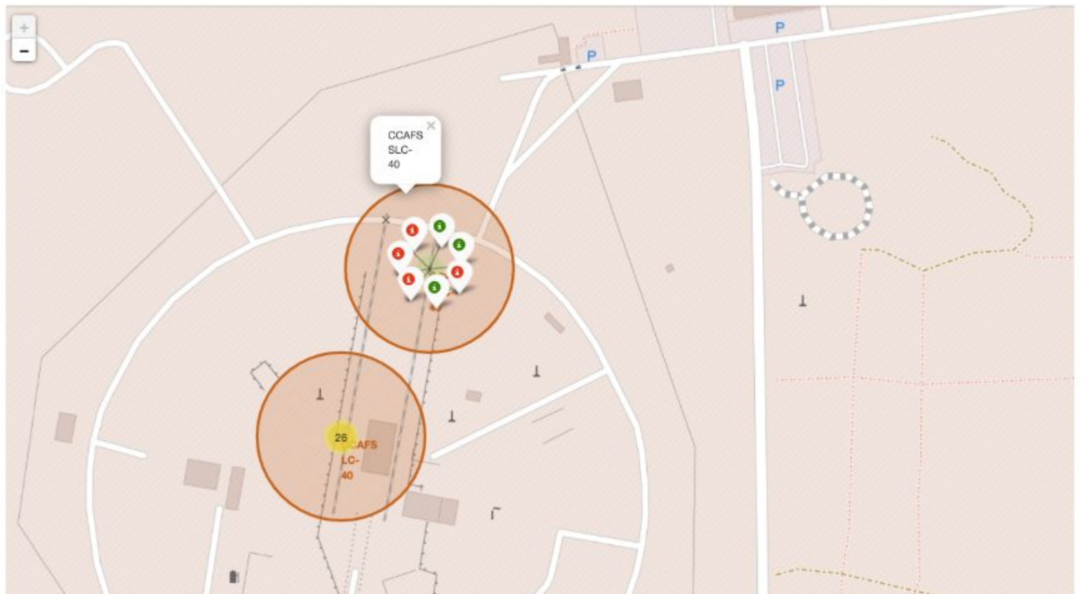
Section 3

Launch Sites Proximities Analysis

Launch locations

Sites with success/failure launches



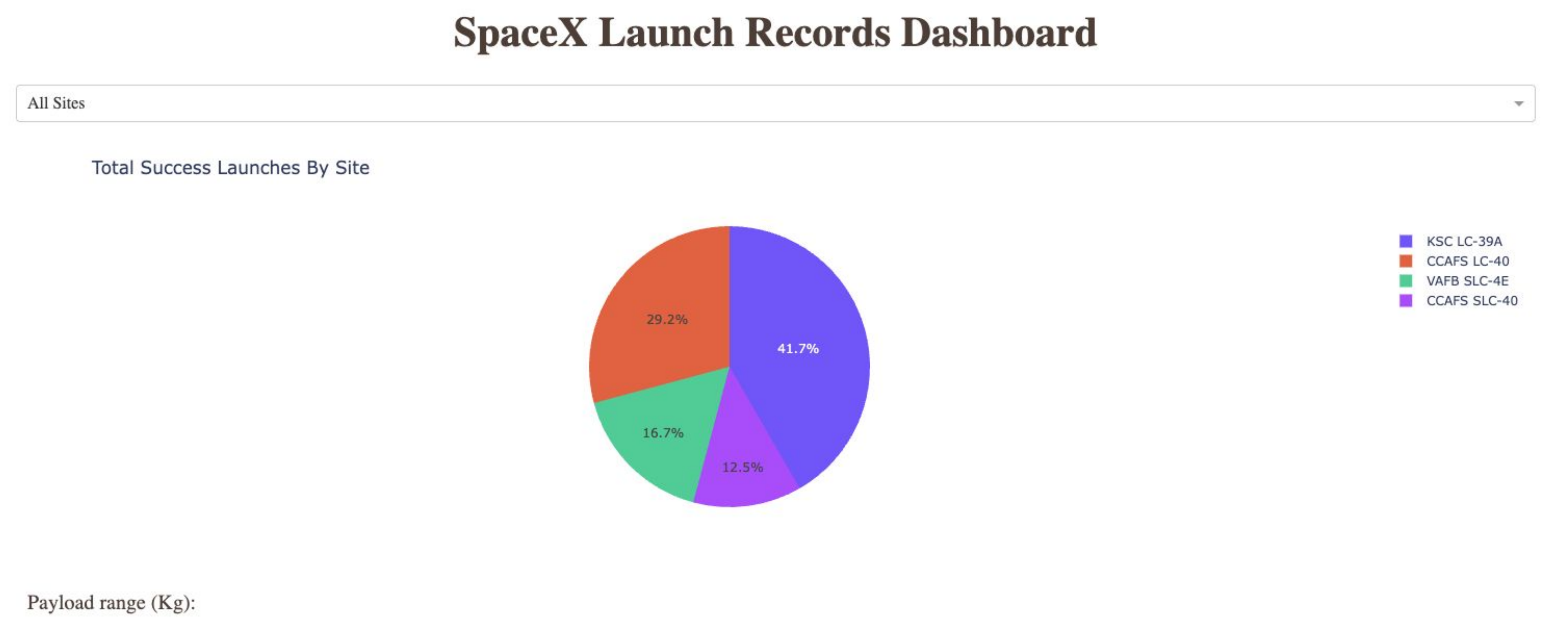


Zoom in the map

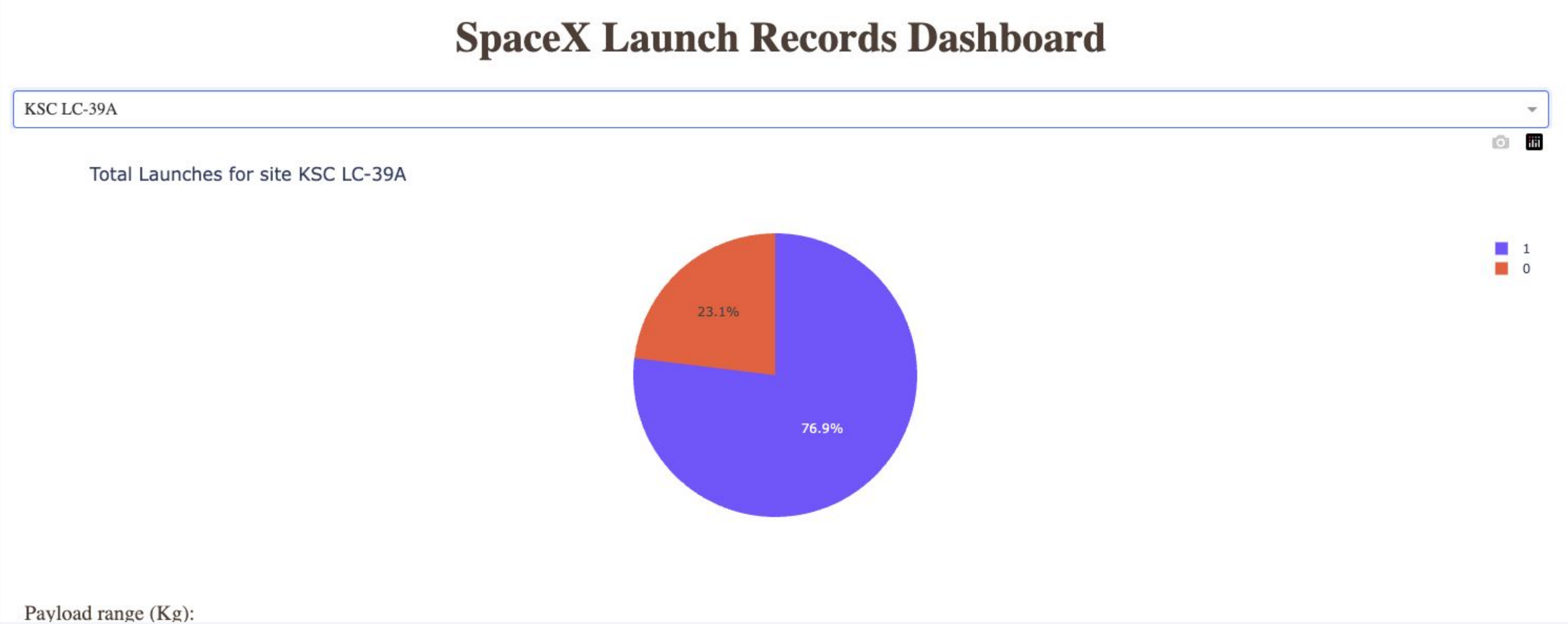
Section 4

Build a Dashboard with Plotly Dash

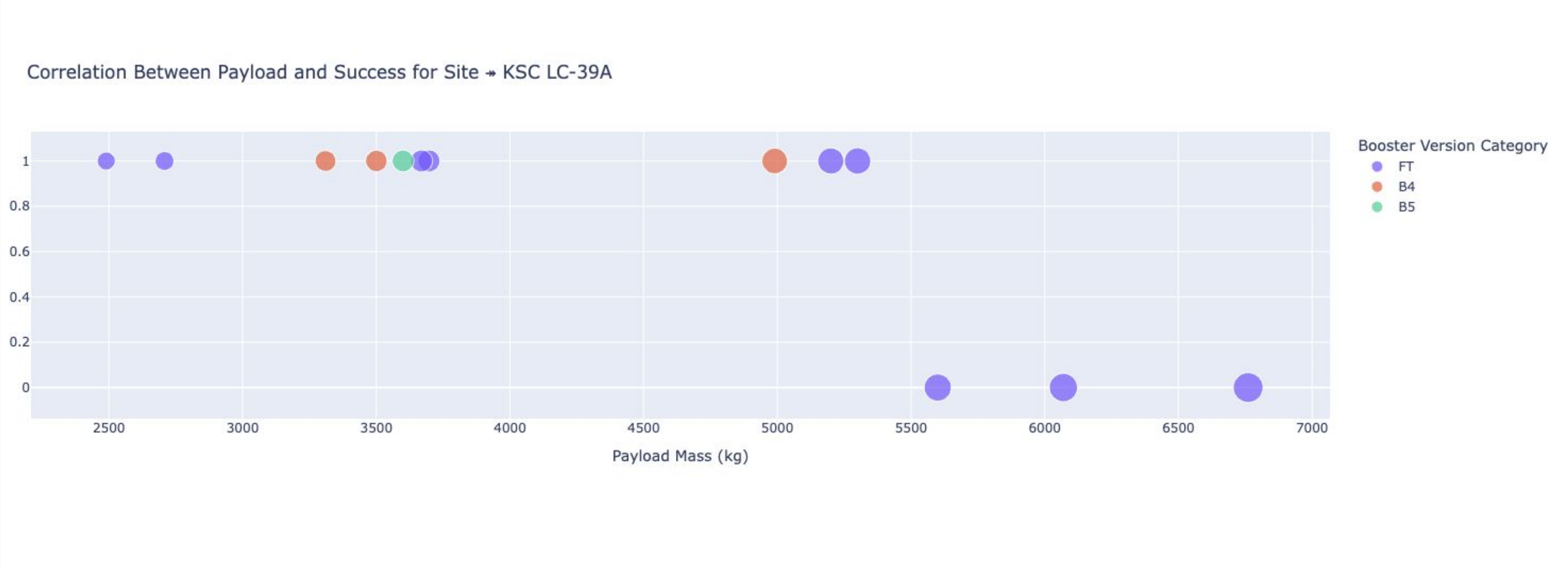
All launch sites



Launch site with highest launch success ratio



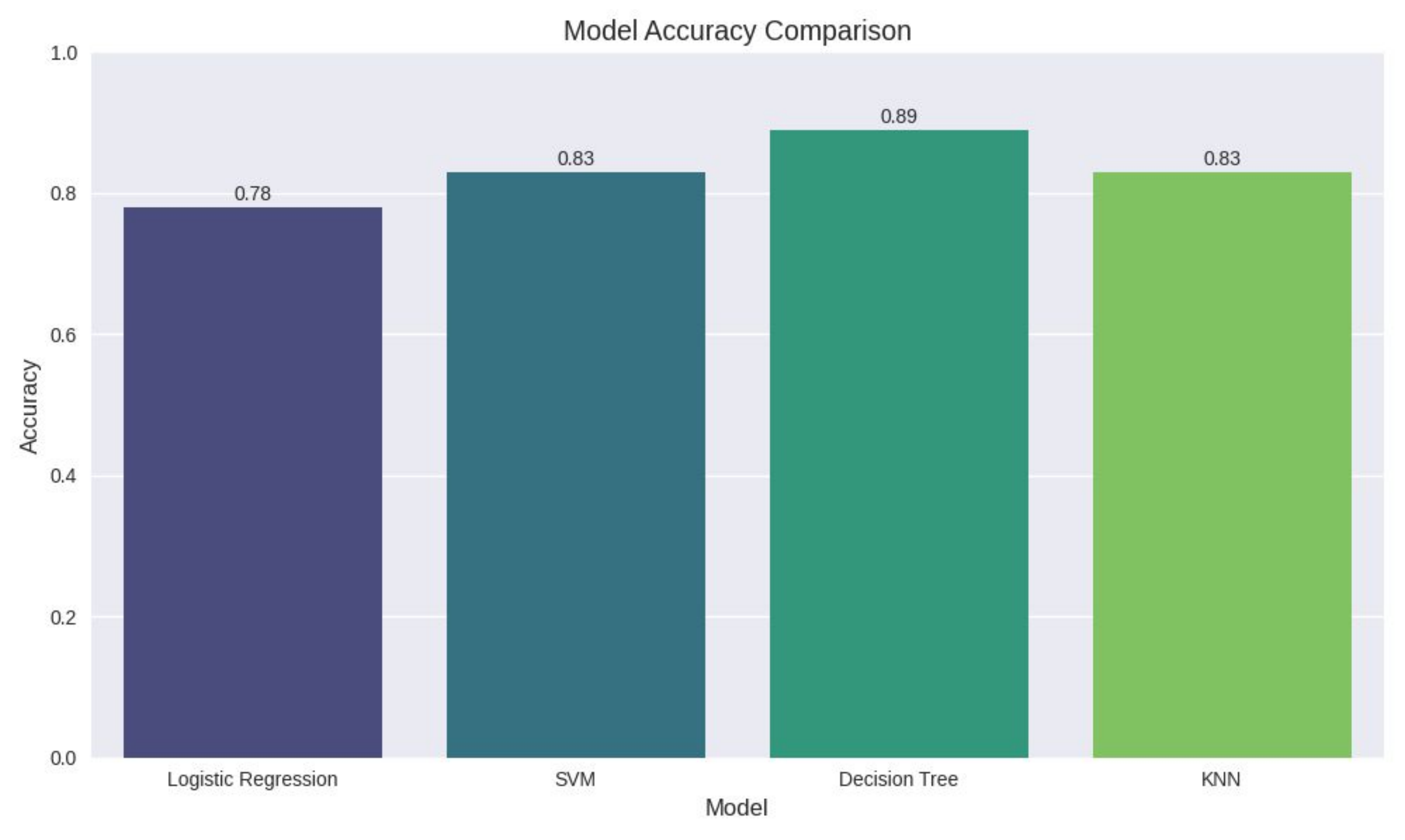
Payload vs. Launch Outcome



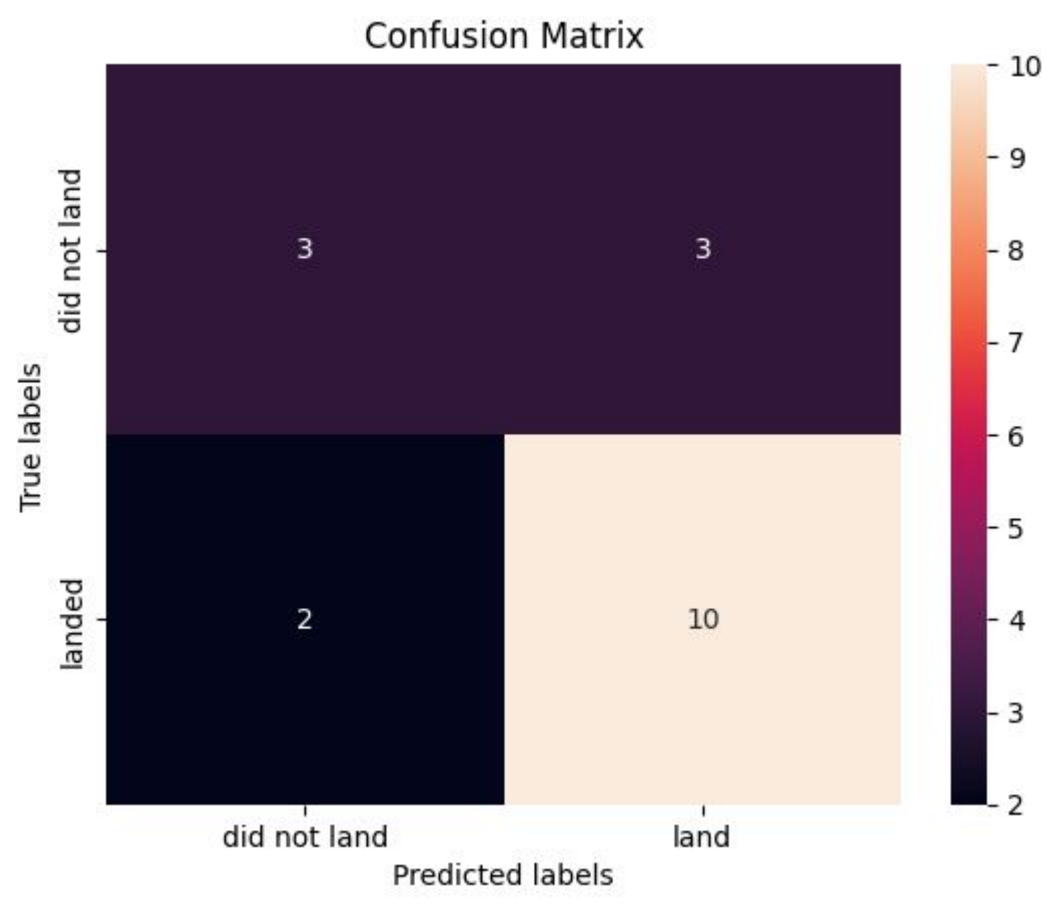
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



Conclusions

- Achieve 89% accuracy with decision tree classifier
- Payload mass and orbit type are significant factors

Appendix

- Data sources
  - SpaceX API: https://api.spacexdata.com/v4/launches
  - Wikipedia: Falcon 9 launch history page
- Repository
  - https://github.com/ittus/learn-data-science/tree/main/capstone

Thank you!